

NGUYEN GIA HUY

Backend Engineer

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PROFESSIONAL SUMMARY

Backend Engineer working with Golang and Python on financial and data-heavy systems. Currently builds core banking features at Vikki Digital Bank, including lending and overdraft logic, account migration, and integrations with external services. Most interested in service design, data consistency, and performance.

TECHNICAL SKILLS

Languages:	Golang, Python, SQL
APIs:	gRPC, gRPC-Gateway, REST, FastAPI
Databases & Messaging:	PostgreSQL, MongoDB, Redis, OpenSearch, Kafka
Service Infrastructure:	Kong Gateway, Keycloak, Docker, Kubernetes, AWS
Observability:	OpenTelemetry, Prometheus, Grafana

WORK EXPERIENCE

Backend Engineer, Core Banking (Middle Software Engineer) 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Integrated Thought Machine Vault Core with external backend services, connecting the core ledger to the bank's operational and transaction workflows.
- Implemented interest accrual and penalty and late-fee calculation for the Lending and Overdraft products as Python Smart Contracts.
- Implemented repayment and settlement processing for both products.
- Built an account migration backend in Golang on AWS ECS and Lambda, with data stored in RDS. Migration time went from **2.5 hours to 27 minutes**.
- Rewrote the Gauge-based simulation framework that tests this logic in Golang; test runs now take **just over 50 seconds instead of 24 minutes**.
- Set up the AWS deployment (EKS, RDS, ECR, ELB) and CI/CD pipeline for a new Vault Core version.

Software Engineer – Backend & Data Services 04/2025 – 03/2026
Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Migrated the search and analytics backend from Elasticsearch to OpenSearch, re-architecting data ingestion and indexing to improve stability, consistency, and query performance.
- Reworked how the batch pipelines read and process data, adding parallel execution where steps were independent; throughput went up by **150%**.
- Built data services on AWS Lambda and ECS, with S3, Athena, and RDS, that transform large-scale utility data for analytics and operations.
- Managed the services' infrastructure with Terraform and AWS SAM and monitored them with CloudWatch and DataDog.

Software Engineer – API Gateway & Identity 04/2024 – 04/2025
VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Integrated Kong Gateway in front of **100+** microservices and maintained its service-level and network-level access rules.
- Built multi-tenant authentication and role-based authorization on Keycloak for **20+** customer organizations.
- Also operated VNPAY Cloud's OpenStack private cloud, with **50+** production Kubernetes clusters managed through Cluster API (provisioning, scaling, zero-downtime upgrades).
- Automated most data center provisioning and configuration work (about **80%**).

Blockchain Developer – Event-Driven Backend

06/2023 – 09/2023

EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Built event-driven backend services that capture, process, and sync on-chain events into off-chain databases and APIs.
- Integrated them with **5+** blockchain networks (Ethereum, BSC, Polygon, Aptos, Sui), with consistent transaction handling and data integrity.
- Wrote and reviewed **15+** smart contracts (Solidity, Move) for DeFi and NFT use cases.

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Built the API layer with gRPC services exposed as REST/JSON through gRPC-Gateway, backed by Kafka, MongoDB, and Redis; the backend handled up to **1.5K** requests per second at 80–100 ms latency.
- Set up multi-cluster failover and observability with OpenTelemetry, Prometheus, Grafana, ELK, and ClickHouse for **99.95%** uptime.

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting backend on LiveKit with OpenAI-based transcription, voice translation, and summaries; designed the translation pipeline and improved its handling of conversation context.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. in Information Technology, Knowledge Engineering

DevOps on AWS Specialization | Coursera